

$$1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \cdots = \frac{\pi^2}{6}$$

Robin Chapman has collected 14 proofs of this result in a paper at his website (<http://www.maths.ex.ac.uk/~rjc/rjc.html>). Some of the proofs rely on mathematics well beyond the level of Math 133, but some of them don't use anything you don't already know—that doesn't mean they are easy, it just means that with a fair bit of effort you should be able to work your way through them. This is an exposition of the third proof, which comes from a note by Boo Rim Choe in the American Mathematical Monthly in 1987.

We need to do a few preliminary calculations. First note that

$$\sum_1^\infty \frac{1}{n^2} = \sum_1^\infty \left(\frac{1}{(2n)^2} + \frac{1}{(2n+1)^2} \right) = \frac{1}{4} \sum_1^\infty \frac{1}{n^2} + \sum_1^\infty \frac{1}{(2n+1)^2} \quad (1)$$

(manipulations justified by absolute convergence) so $\sum \frac{1}{n^2} = \frac{4}{3} \sum_1^\infty \frac{1}{(2n+1)^2}$, and it

suffices to prove $\sum_1^\infty \frac{1}{(2n+1)^2} = \frac{\pi^2}{8}$.

Next let $I_{2n+1} = \int_0^{\pi/2} \sin^{2n+1} x \, dx$. Note that $I_1 = 1$, and for $n > 0$ we can use integration by parts with $u = \sin^{2n} x$, $dv = \sin x \, dx$, to get

$$\begin{aligned} I_{2n+1} &= -\sin^{2n} x \cos x \Big|_0^{\pi/2} + 2n \int_0^{\pi/2} \sin^{2n-1} x \cos^2 x \, dx \\ &= 2n \int_0^{\pi/2} \sin^{2n-1} x (1 - \sin^2 x) \, dx = 2nI_{2n-1} - 2nI_{2n+1} \end{aligned}$$

This rearranges to $I_{2n+1} = (2n/(2n+1))I_{2n-1}$, so it reduces the problem of computing I_{2n+1} to that of computing I_{2n-1} . But then you can apply it again, and again, and again, until eventually it reduces to calculating I_1 , which is already done. You get

$$I_{2n+1} = \frac{2 \times 4 \times \cdots \times 2n}{3 \times 5 \times \cdots \times (2n+1)} \quad (2)$$

We also need a formula for the binomial coefficient $\binom{-1/2}{n}$.

$$\binom{-1/2}{n} = \frac{(-1/2)(-3/2)(-5/2) \cdots (-(2n-1)/2)}{(1)(2)(3) \cdots (n)} = (-1)^n \frac{1 \times 3 \times 5 \times \cdots \times (2n-1)}{2 \times 4 \times 6 \times \cdots \times (2n)} \quad (3)$$

Finally, we need the Maclaurin series for $\arcsin x$. We have

$$\frac{d}{dx} \arcsin x = (1-x^2)^{-1/2} = \sum_0^\infty \binom{-1/2}{n} (-x^2)^n$$

$$\sum \frac{1}{n^2} = \frac{\pi^2}{6}$$

for $|x| < 1$ by the binomial theorem, and antidifferentiating we get

$$\arcsin x = \sum_0^{\infty} \binom{-1/2}{n} (-1)^n \frac{x^{2n+1}}{2n+1} \quad (4)$$

Here and again later we omit the justification for term-by-term antidifferentiation of an infinite series. Also, we assert without proof that (4) converges for $|x| \leq 1$.

Now we can give the proof. In (4), let $x = \sin t$;

$$t = \sum_0^{\infty} \binom{-1/2}{n} (-1)^n \frac{\sin^{2n+1} t}{2n+1} \quad (5)$$

Integrate both sides of (5) from 0 to $\pi/2$;

$$\frac{\pi^2}{8} = \sum_0^{\infty} \binom{-1/2}{n} (-1)^n \frac{1}{2n+1} \int_0^{\pi/2} \sin^{2n+1} t \, dt$$

Now use (3) and (2) to get

$$\frac{\pi^2}{8} = \sum_0^{\infty} \frac{1 \times 3 \times 5 \times \cdots \times (2n-1)}{2 \times 4 \times 6 \times \cdots \times (2n)} \frac{1}{2n+1} \frac{2 \times 4 \times \cdots \times 2n}{3 \times 5 \times \cdots \times (2n+1)} = \sum_1^{\infty} \frac{1}{(2n+1)^2}$$

and we're done.